Reporting Regulation, Stock Liquidity, and International Diffusion

Abstract

We examine reporting regulation across countries using stock-market data from 90 jurisdictions over 2000-2025. Seventeen policy families are considered; 15 support first-year liquidity comparisons and nine support a balanced longer window. We estimate changes around recorded national implementation dates relative to comparison jurisdictions, then aggregate across regulation families. First-year log-spread estimates are -0.029 with equal family weights and +0.021 with market-capitalisation weights. The market-cap estimate is imprecise, and the liquidity measures do not establish a uniform improvement. We also examine international diffusion using lagged competition, learning, emulation and external-finance measures. No association survives correction across the complete model matrix. The analysis describes exposure to national policy changes; it does not establish the reporting obligations of every company.

1. Introduction

Disclosure regulation can give investors more information and make companies easier to compare. If it narrows information differences between traders, shares may become easier to trade. Governments have extended reporting beyond financial statements to nonfinancial performance, ownership, supply chains and payments to governments. We ask whether liquidity changes accompany this wider set of reforms and how the policies spread across countries.

We extend the international disclosure setting in Krueger et al. (2023) to a wider set of regulation families. National implementation dates are linked to company stock outcomes to examine liquidity before and after implementation. We then examine international adoption using the diffusion framework of Koning, Mertens and Roosenboom (2018). Original company reports provide supplementary evidence on reporting scope and timing.

The policies differ in what information they provide and who receives it. Public country reports can inform investors; confidential country-by-country tax reports go to tax authorities. Audit institutions, governance revisions and sectoral rules may affect markets without creating a new disclosure duty for every listed company. An average across these policies describes a broad policy environment rather than one common reporting treatment.

2. Sample and research design

Datastream supplies 1,055,142 stock-year observations in 90 jurisdictions from 2000 to 2025, linked to 99,100 vendor company records. One stock-year is one stock observed for one year. Each liquidity measure requires at least 120 valid daily observations. We use the log relative quoted spread, log Amihud price impact and fraction of observed returns equal to zero; higher values indicate lower liquidity. Logged measures are winsorised each year at the first and ninety-ninth percentiles.

Table 1 reports all 17 attempted families. The first-year comparisons cover 30,703 distinct treated company records in 48 jurisdictions across 15 families. The longer-window comparisons cover 18,470 treated company records in 36 jurisdictions across nine families. Companies may appear under several policies; company-policy exposures are not unique companies. Some families have broad international support, while others have only one supported treated jurisdiction.

For each regulation family and implementation cohort, we compare the change from the preceding year in implementing jurisdictions with the change in comparison jurisdictions. Already implementing countries do not serve as controls for that family. Comparisons use documented later dates; the bank home-jurisdiction contrast also includes explicitly coded non-EU jurisdictions, which may still have national rules or affected subsidiaries. Missing histories are not coded as nonadoption. National implementation is an exposure assignment, not proof that each company must report.

We select one share line per company using the largest market value before implementation and keep it fixed within each cohort. The first-year design uses years -1 and 0. The longer window requires complete outcomes from -3 through +2 and averages years 0-2 relative to -1. Each country requires 30 complete companies; each comparison requires 30 treated companies, 30 comparison companies and three comparison jurisdictions. The narrower longer-window sample is not a continuation of the first-year aggregate.

Within a family, prior-year country GDP is divided across its complete company sample. Across families, equal weights describe the average supported family; market-cap weights use the sampled treated companies' pre-implementation market value. These are exposure weights across country-policy cohorts, not shares of worldwide legally covered issuers. Company exposure can recur across families. Country-clustered intervals condition on the observed implementation calendar; collective legal waves and few treated countries can make policy-level uncertainty larger. Date-plus-one and independent-national-choice specifications remain in the supplement.

3. Liquidity

Table 2 reports the international estimates under both family weights. First-year log spreads are -0.029 under equal weights and +0.021 under market-cap weights. Equal-weight price impact is -0.045 and the unchanged-return fraction is -0.0082; the corresponding market-cap estimates are +0.019 and -0.0027. Only the equal-weight unchanged-return result survives the adjustment across the 12 principal liquidity comparisons. The evidence does not establish lower trading costs across regulation families.

Figure 1 reports the family estimates and their weights. Confidential country-by-country tax reporting receives 33.2% of first-year market-cap weight; international auditor reporting receives 9.9%. These weights reflect the global sampled exposure population. Figure 2 reports the balanced nine-family profiles. Pre-implementation differences are material, so subsequent changes cannot be attributed to the policies alone. Overlapping reforms, sector-specific eligibility and prior voluntary disclosures remain alternative explanations.

4. International diffusion

We examine whether implementation follows earlier adoption among trade peers, earlier economic outcomes among geographic neighbours, language or institutional peers, and external financial dependence. Signals precede the focal decision. Table 3 reports four pooled timing regressions with family-year strata and country-clustered uncertainty. The samples differ because the required histories differ. The external-finance coefficient is negative in this narrow comparison; its adjusted p-value is 0.0013 across these four models but 0.142 across the complete stable-coefficient matrix.

The full exercise contains 1,330 timing and adoption-status designs and 433 stable focal coefficients. None survives the full multiplicity correction. These associations do not establish that one country caused another to adopt. Together, the liquidity and diffusion analyses provide international evidence on recorded policy implementation while distinguishing policy exposure, company eligibility and the information actually available to investors.

References

Krueger, P., Z. Sautner, D. Y. Tang and R. Zhong. 2023. The Effects of Mandatory ESG Disclosure Around the World. Supplied working paper.

Koning, M., G. Mertens and P. Roosenboom. 2018. Drivers of institutional change around the world: The case of IFRS. Journal of International Business Studies. Supplied article.

Policy chronology, official source links, legal-scope reviews, complete estimates and implementation sensitivities: https://hunternbh.github.io/policy-diffusion/ . Licensed Datastream observations remain local.

Table 1. International policy coverage

Regulation family	Treated countries	Supported cohort years	Company-policy cohort exposures	Market-cap weight	Longer window
Audit institutions / rotation	4	2004-2007	3,399	2.6%	Yes
Auditor reporting	19	2012-2016	5,364	9.9%	Yes
Beneficial ownership	11	2016-2020	6,507	10.3%	Yes
Public bank country reports	15	2014	3,288	6.5%	Yes
Confidential tax country reports	42	2016-2020	21,796	33.2%	Yes
Public multinational country reports	1	2023	32	<0.1%	No
Climate / emissions disclosure	2	2013-2023	711	2.0%	Yes
Sectoral climate rules	-	-	-	-	No
Digital-asset regulation	2	2018-2020	762	0.5%	Yes
Nonfinancial reporting	23	2002-2019	8,265	10.7%	Yes
Payments to governments	5	2014-2015	2,767	6.4%	No
Corporate governance	6	2001-2004	2,385	1.2%	No
IFRS financial reporting	21	2005-2011	5,841	6.5%	No
Supply-chain reporting	5	2012-2022	5,012	9.5%	Yes
Financial-focus sustainability	3	2024-2025	713	0.5%	No
Sustainability frameworks	-	-	-	-	No
Electronic financial reporting	1	2007	345	0.2%	No

First-year GDP-weighted specification. Only supported implementation cohorts through 2025 appear; future schedules are excluded. Counts refer to national policy exposure, not verified legally covered companies. No means insufficient longer-window support, not a zero effect. Public bank phases enter once. Audit institutions/rotation and governance frameworks include different policy subtypes. Nonfinancial dates are reviewed reporting phases, not a harmonised claim to the earliest E, S and G duties in every country.

Scope and timing

The calendar records the implemented policy phase supported by the source register. It does not claim the first existence of every reporting obligation. E-only climate reporting is distinct from comprehensive nonfinancial reporting; public and confidential country reports have different audiences. Historical thresholds, subsidiaries and foreign-listing exemptions are not fully observed. Unknown country histories remain outside the comparison set. The two unsupported families remain in the inventory without an estimated effect.

Table 2. International liquidity comparisons

Window and outcome	Equal family Estimate [interval]	Market-cap family Estimate [interval]
First year: 15 families Bid-ask spread (log)	-0.029 [-0.068, +0.011]	+0.021 [-0.021, +0.063]
First year: 15 families Price impact (log)	-0.045 [-0.087, -0.003]	+0.019 [-0.050, +0.087]
First year: 15 families Unchanged returns (fraction)	-0.0082 [-0.0112, -0.0052]	-0.0027 [-0.0065, +0.0012]
Years 0-2: 9 families Bid-ask spread (log)	-0.017 [-0.078, +0.044]	+0.045 [-0.035, +0.125]
Years 0-2: 9 families Price impact (log)	-0.013 [-0.081, +0.054]	+0.096 [-0.024, +0.216]
Years 0-2: 9 families Unchanged returns (fraction)	-0.0006 [-0.0064, +0.0053]	+0.0070 [-0.0070, +0.0210]

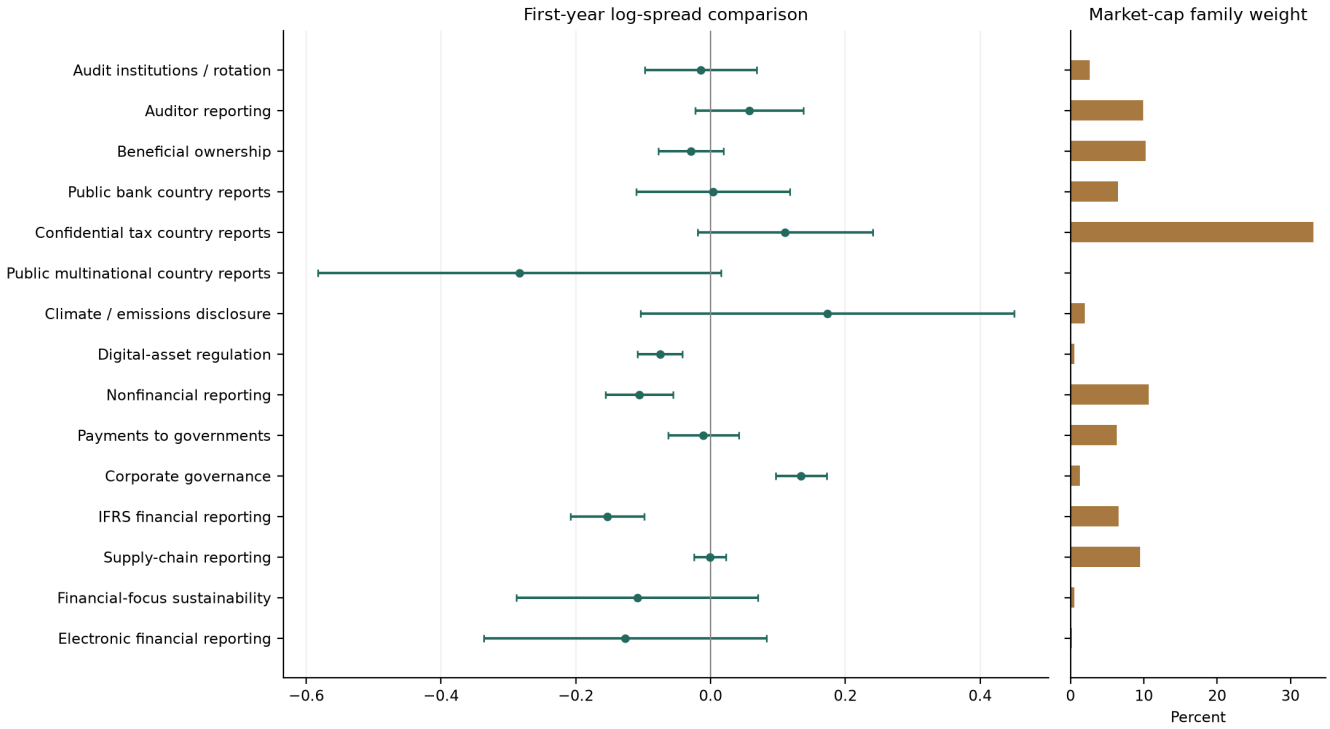
GDP weights within regulation families. Brackets are 95% country-clustered conditional intervals; they do not capture all uncertainty from collective policy waves, scope errors or endogenous implementation. Higher outcomes indicate lower liquidity. Log outcomes and fractions are not percentage changes. Horizons have different family and company support. All outcomes and both weights are reported.

Table 3. International diffusion

Mechanism	Rows / countries	Logit coefficient	Raw p	Full-matrix adjusted p
Competition	233 / 58	+0.458	0.0883	1.000
Learning	74 / 33	-0.409	0.5601	1.000
Emulation	411 / 66	+0.105	0.7032	1.000
External finance	269 / 48	-0.720	0.0003	0.142

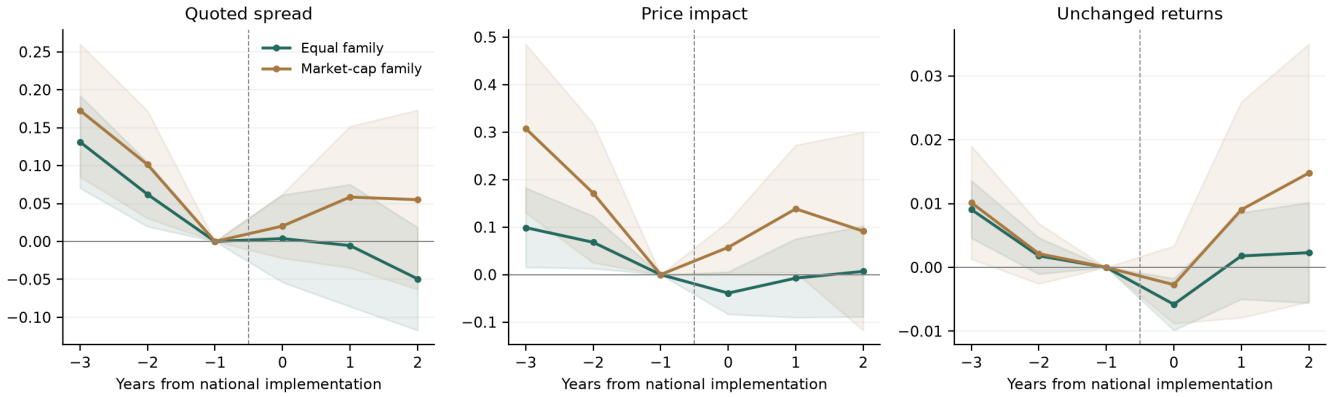
Pooled first-recorded-implementation timing; separate single-mechanism logit models, family-year strata, country clustering. Continuous predictors are standardised within each model. Competition uses fixed-2000 export-pattern peers; learning uses lagged neighbouring GDP outcomes; emulation uses language peers; external finance combines credit and aid measures. These are associations in documented risk sets. Full-matrix Holm correction covers all 433 stable focal coefficients; the four-model correction gives 0.0013 for external finance.

Figure 1. International family estimates and weights



First-year national policy-exposure comparisons for all 15 supported families, GDP weighted within families. Bars show country-clustered conditional intervals. Right panel: sampled treated-company market capitalisation before implementation.

Figure 2. Liquidity around national implementation



Nine families with fixed -3 through +2 company support. Year -1 is the reference; year 0 is recorded national implementation. Shading shows pointwise country-clustered conditional intervals. Pre-implementation differences and collective legal waves limit a causal interpretation. This sample differs from the first-year sample.